Deep Reinforcement Learning for AoI-Aware UAV-Assisted Networks With RF Energy Harvesting

Gaurav Kumar Pandey, *Graduate Student Member, IEEE*, Devendra Singh Gurjar, *Senior Member, IEEE*, Suneel Yadav, *Senior Member, IEEE*, and Xingwang Li, *Senior Member, IEEE*

***Abstract*—This letter considers UAV-assisted data collection from energy-constrained Internet of Things (IoT) devices. Herein, a UAV utilizes radio frequency-based wireless power transfer technique to charge multiple IoT devices or schedules one IoT device to transmit its sensed data. Using the harvested energy, the IoT devices share the collected data with the UAV as per their schedule. For this setup, we aim to minimize IoT devices' average Age of Information (AoI) by optimally controlling the UAV's trajectory and scheduling of IoT devices while adhering to the energy consumption limitations of UAV and IoT devices. Considering the dynamic scenario for the considered network, the optimization problem is modeled as a Markov Decision Process and solved through dueling double deep Q-networks (D3QN) algorithm. The simulation results show that the proposed framework outperforms the baseline methods in reducing the average AoI of the IoT devices.**

***Index Terms*—Uncrewed aerial vehicle, age of information, UAV-assisted data collection, deep reinforcement learning.**

I. Introduction

THE INTERNET of Things (IoT) can play a crucial role in supporting human-centric and mission-driven applications for the upcoming sixth-generation communications. Large-scale IoT devices are deployed to monitor various physical processes like temperature, light intensity, etc., and periodically update the data. However, the limited communication range and low power of IoT devices make it challenging for them to transmit data to distant data centers. In such situations, uncrewed aerial vehicles (UAVs) can be utilized as they maintain a strong line-of-sight (LoS) link and are easy to deploy [1]. These UAV-enabled IoT networks are particularly suited for delay-sensitive applications like traffic management, autonomous driving, etc. In these applications, the freshness of collected data is crucial for making timely decisions, which can be effectively quantified through a metric termed as Age of Information (AoI) [2]. It refers to the time that has elapsed since the recent packet was received at the destination and generated by IoT devices. Reducing the AoI in UAV-collected data is essential to ensure data validity for quick decision-making responses. Additionally, UAVs can wirelessly charge IoT devices while simultaneously gathering data from them [3], thereby prolonging the IoT network's operational time.

Various researchers have explored the data freshness in UAV-assisted data collection in IoT networks. Specifically, the authors in [4] utilized successive convex approximation and genetic algorithms to design an optimal AoI-energy-aware trajectory of UAVs. However, they rely on heuristic algorithms, which do not guarantee optimal solutions. Further, Zhu et al. optimized the hover point selection and access sequence for the UAVs through a machine learning-based path planning algorithm [5]. In another study, Fu et al. focused on energy-AoI-aware UAV-assisted data collection and enabled air-to-air transmissions between UAVs to address mission allocation challenges [6]. In contrast, the authors in [7] and [8] explored deep reinforcement learning (DRL)-based minimal AoI strategies for UAV-based data collection. Notably, these works [4], [5], [7], [8] did not incorporate UAV-assisted RF-WPT in their models, whereas the authors in [6] assumed infinite battery capacity at the EH nodes, limiting their real-world applicability.

Motivated by the above works, we propose a UAV-assisted AoI-aware data collection scheme for multiple energy-constrained IoT devices deployed within a geographic area. A UAV is assigned to wirelessly charge the IoT devices through RF-based WPT and collect the data they have sensed. At any given time, the UAV either charges the IoT devices or schedules an IoT device to transmit its data. To realize a more practical scenario, we use the LoS-probabilistic model and Rician fading to characterize the channel between the UAV and IoT devices. For this setup, we minimize the average AoI (AAoI) by jointly optimizing the UAV's trajectory and scheduling the IoT device for data collection or WPT. Due to the system's dynamic nature, we reformulate the problem as a Markov Decision Process (MDP) with large state spaces. We propose a deep reinforcement learning (DRL) approach to solve it efficiently using a dueling double deep Q-network (D3QN) algorithm. Lastly, we provide the simulation results, demonstrating that the proposed D3QN-based UAV-assisted data collection scheme outperforms the baseline schemes by achieving the lowest AAoI.

II. System Model Discussion

A. System Model

We consider a UAV-assisted IoT network with a set of IoT devices, denoted as $\mathcal{D} = \{D_1, D_2, \ldots, D_N\}$. Each IoT

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Gaurav Kumar Pandey and Devendra Singh Gurjar are with the Department of Electronics and Communication Engineering, National Institute of Technology Silchar, Silchar 788010, India (e-mail: gaurav_rs@ece.nits.ac.in; dsgurjar@ece.nits.ac.in).

Suneel Yadav is with the Department of Electronics and Communication Engineering, Indian Institute of Information Technology Allahabad, Prayagraj 211015, India (e-mail: suneel@iiita.ac.in).

Xingwang Li is with the School of Physics and Electronic Information Engineering, Henan Polytechnic University, Jiaozuo 454000, China (e-mail: lixingwang@hpu.edu.cn).

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device, D_n for $n \in \{1, 2, \ldots, N\}$, is deployed in a specific geographical area to sense and collect the environmental parameters. Due to their limited battery capacity, these IoT devices often suffer from energy depletion and thus depend on the UAV, referred to as U, for energy supply through RF-WPT. These IoT devices utilize EH and store the energy in their buffer with a maximum capacity of $B_n^{\max}$ J, which is utilized to their sensed data to U. Moreover, U either charges all IoT devices or schedules D_n to upload its sensed data during the lth time slot once it has accumulated sufficient energy for transmission. All nodes are assumed to be equipped with a single antenna and operate in half-duplex mode.

In a Cartesian coordinate system, each D_n's horizontal position is given by $\mathbf{q}_n = [x_n, y_n]$. The UAV aims to charge all IoT devices and collect their sensed data within a finite time interval T, which is divided into L equal time slots, each with a duration of $\Lambda = \frac{T}{L}$. Moreover, U's position can be considered stationary by selecting larger values of L [9]. Thus, the U's projected horizontal position in the lth time slot is $\mathbf{q}_U(l) = [x_U(l), y_U(l)]$ and its three-dimensional position is given by $(\mathbf{q}_U(l), H)$, where H is the fixed altitude of U. It begins its flight at the time-varying velocity ν_U (l) from the horizontal position $\mathbf{q}_0 = [x_0, y_0]$ and ends at $\mathbf{q}_F = [x_F, y_F]$. For designing the U's trajectory, we consider a flight area having a grid size of $\mathcal{D}_x \times \mathcal{D}_y$ squares, such that $\mathcal{D}_x \ll L$ and $\mathcal{D}_y \ll L$. Moreover, U is restricted to moving one grid per time slot, i.e., $\nu_U(l) \times \Lambda$ does not exceed the grid size, resulting in a smoother trajectory.

B. Channel Model

The presence of obstacles such as buildings, tall trees, or mountains can severely hinder the nodes' direct links, resulting in non-LoS (NLoS) connectivity in UAV-assisted communications. Thus, it is essential to consider the randomness associated with the LoS and NLoS links while modeling the channel [10]. For data collection via a ground-to-air link, D_n has an LoS connection to U's specific locations with a probability that depends on the elevation angle between the D_n and U and operating scenario. The elevation angle (in degrees) is given as $\theta_{nU}(l) = \frac{180}{\pi}\sin^{-1}(\frac{H}{d_{nU}(l)})$, where $d_{nU}(l) = \sqrt{\|q_U(l) - q_n\|^2 + H^2}$ is the distance between U and D_n. Therefore, the LoS probability between U and D_n in the lth time slot is denoted as $P_{nU}^{\mathrm{LoS}}(l) = \frac{1}{1+A_1\exp(-A_2\theta_{nU}(l)-A_1)}$, where A_1 and A_2 are environment-dependent parameters, which varies based on the operating environment, such as rural, urban, or dense urban. From the expression of $P_{nU}^{\mathrm{LoS}}(l)$, we observe that an increasing elevation angle results in a larger LoS probability. Consequently, the NLoS probability is obtained as $P_{nU}^{\mathrm{NLoS}}(l) = 1 - P_{nU}^{\mathrm{LoS}}(l)$. The channel power gain g_{nU} between D_n and U, based on LoS and NLoS, is given as

$$g_{nU} = \left(\frac{1}{\eta_1}P_{nU}^{\mathrm{LoS}}(l) + \frac{1}{\eta_2}P_{nU}^{\mathrm{NLoS}}(l)\right)\beta_0|\tilde{h}_{nU}|^2(d_{nU})^{-\alpha}. \quad (1)$$

The path-loss exponent is denoted as α, and η_1 and η_2, where $\eta_2 > \eta_1 > 1$ are the excessive path-loss coefficients. Moreover, β_0 denotes the channel power gain coefficient at a unit distance of 1 m. The small-scale fading coefficients between U and D_n are denoted as $\{\tilde{h}_{nU}\}$ and is characterized using Rician fading distribution, given as $\tilde{h}_{nU}(l) = \sqrt{\frac{K(l)}{K(l)+1}} + \sqrt{\frac{1}{K(l)+1}}h_{nU}$, where $K(l)$ is the Rician factor and $h_{nU} \sim \mathcal{CN}(0, 1)$.

C. Energy Harvesting and Transmission Model

The UAV broadcasts RF signals to the IoT devices in the lth time slot, and signal power received at D_n is given as $P_n(l) = g_{U,n}(l)P_U$, where P_U is the transmit power of U. Using a piecewise non-linear EH model, the harvested energy by D_n in time slot l is expressed as

$$E_n^H(l) = \begin{cases} \eta\Lambda g_{U,n}(l)P_U, & \text{if } P_n(l) \le P_n^{\mathrm{sat}} \\ \eta\Lambda P_n^{\mathrm{sat}}, & \text{if } P_n(l) > P_n^{\mathrm{sat}} \end{cases} \quad (2)$$

where η is the energy conversion efficiency and P_n^{sat} is the saturation threshold of the EH receiver of D_n. If D_n is scheduled by U in the lth time slot, it generates and uploads a data packet of length D_n immediately. According to Shannon's formula [11], the energy required for D_n to successfully transmit data in slot l can be expressed as $E_n^T(l) = \Lambda(\frac{2^{\frac{L_n}{\Lambda B}}-1}{g_{nU}(l)})N_U$, where B is channel bandwidth, N_U denotes noise power at U, and L_n is packet length. Therefore, the signal-to-noise ratio for $D_n - U$ links are expressed as $\gamma_{\mathrm{n,U}}(l) = \frac{P_n^H g_{nU}(l)}{N_U}$, where P_n^H represents the power available at the D_n through EH. It is crucial that the sensed data is successfully transmitted to U, i.e., $\gamma_{\mathrm{n,U}}(l) \ge \gamma_{\mathrm{th}}$. Since U requires a significant amount of power for propulsion and follows its trajectory, the aerodynamic power consumption of U must be considered [2], which comprises two major components, i.e., induced power and blade profile power associated with the blade drag. Therefore, the aerodynamic energy consumption of U in l-slot, indicated as $E_{\mathrm{U}}^{\mathrm{aer}}(l)$, is expressed as

$$E_{\mathrm{U}}^{\mathrm{aer}}(l) = \left(P_0\left(1 + \frac{3\nu_U^2(l)}{\mathcal{S}_{\mathrm{tip}}^2}\right) + P_i v_0 \nu_U(l) + \frac{1}{2}d_0\mu s A\nu_U^3(l)\right)\Lambda, \quad (3)$$

where blade profile power and induced power parameters in hovering state are represented by P_0 and P_i, respectively. The tip speed of rotor blade is indicated by $\mathcal{S}_{\mathrm{tip}}$, while v_0 refers to the mean induced velocity of rotor during hovering. The fuselage drag ratio is denoted by d_0, s represents the rotor solidity, μ is the air density, and A refers to rotor disc area.

D. Age of Information (AoI) Model

We define the AoI metric to assess the freshness of data collected by U, which is the time elapsed since the IoT device generated the most recent information. In the lth time slot, the AoI of data collected from D_n is expressed as $A_n(l) = (l - V_n(l))$, where $V_n(l)$ denotes the time when U received the information from D_n previously. Importantly, D_n needs to be scheduled at least once within T to ensure the data freshness. However, if it is scheduled multiple times, the latest transmitted data only determines the AoI. We introduce $b(l)=\{0, 1, \ldots, K\}$, which indicates the transmission scheduling of IoT devices, i.e., $b(l) = 0$ denotes that all IoT devices are scheduled to harvest the energy, whereas $b(l) =$

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$k, \forall k \in \{1, 2, \ldots, K\}$ indicating D_n's schedule to upload the data. Therefore, the AoI of data collected in $(l+1)$-time slot is expressed as

$$A_n(l+1) = \begin{cases} 1, & b(l) = k, \gamma_{\mathrm{n,U}}(l) \geq \gamma_{\mathrm{th}}, E_n(l) \geq E_n^T(l) \\ \min(A_n^{\max},\ A_n(l)+1), & \text{otherwise} \end{cases} \tag{4}$$

where $E_n(l)$ is the energy level of D_n's battery. In (4), the AoI of D_n's data, denoted by $A_n(l+1)$, evolves in the following way, i.e., when the scheduled D_n uploads its data, AoI resets to 1; or else, it increases by 1 in each time slot, reaching to a maximum tolerable AoI value, denoted as $A_n^{\max}$.

III. Problem Formulation

The proposed work aim to minimize AAoI at U by jointly optimizing the UAV's trajectory and scheduling of UAV-assisted WPT or data upload from IoT device. Based on the discussion, AAoI minimization problem is formulated as

$$(\mathrm{P1}): \quad \min_{\mathcal{A},\mathcal{Q}} \frac{1}{L}\sum_{l=1}^{L} A_n(L) \tag{5a}$$

$$\textbf{st.}\ a_n(l)E_n^T(l) \leq E_n(l), \tag{5b}$$

$$a_w(l)E_n^H(l) + E_n^H(l-1) \leq B_n^{\max}, \tag{5c}$$

$$\sum_{l=1}^{L} E_U^{\mathrm{aer}}(l) + P_U L \leq B_U^{\max}, \tag{5d}$$

$$\sum_{n=1}^{N} a_n(l) + a_w(l) = 1, \tag{5e}$$

$$a_n(l) \in \{0,1\}, \quad a_w(l) \in \{0,1\}, \tag{5f}$$

$$\mathbf{q}_U(0) = \mathbf{q}_0, \quad \mathbf{q}_U(N) = \mathbf{q}_F, \tag{5g}$$

$$\mathbf{q}_U(l) - \mathbf{q}_U(l-1) \leq \nu_U(l)\ \Lambda, \tag{5h}$$

$$\|\nu_U(l)\| \leq \nu_U^{\max}, \tag{5i}$$

where $A_n(L)$ can be obtained by using (4). Assuming, $\mathcal{A} = [a(0), \ldots, a(N)]$ with $a(l) = \{a_1(l), \ldots, a_N(l), a_w(l)\}$ indicate IoT devices' schedule vector in the lth time slot and $\mathcal{Q} = [\mathbf{q}_U(0), \ldots, \mathbf{q}_U(L)]$ to be U's trajectory vector. If $a_n(l) = 1$, D_n is scheduled for data upload in the lth slot; otherwise, no information transfer takes place. Whereas the IoT devices are scheduled for EH when $a_w(l) = 1$ in the lth slot; otherwise, no WPT occurs from U. Constraints are defined as (5b) ensures that the energy used by D_n does not exceed the stored energy. Whereas (5c) limits the harvested energy over l-slots to the maximum battery capacity $B_n^{\max}$. Eq. (5d) restricts the total energy used by the UAV to be within $B_U^{\max}$. Eq. (5e) and (5f) ensure that all IoT devices are either scheduled for EH or D_n is scheduled for data upload in the lth slot. Eq. (5g) specifies that the UAV must start from the initial location and reach the final destination by T. Moreover, (5h) and (5i) are velocity restrictions, where U can adjust its velocity from one slot to the next slot at most, with $\nu_U^{\max}$ denoting the maximum allowable U's velocity. It can be observed that ($P1$) is a nonlinear integer programming problem with non-convex constraints, and the coupling of binary variables $\mathcal{A}$ and $\mathcal{Q}$ makes it difficult to tackle with traditional methods. In the next section, (P1) is modeled as MDP problem. Then, a D3QN-based UAV-assisted fresh data collection scheme is proposed which structures two networks, i.e., one to evaluate the reward and other to predict the most promising action.

IV. DRL-Based AAoI Minimization Approach

A. Formulated MDP Problem

The fundamental components, states (s), actions (α), reward (ϱ), and state transition probability (ρ) are typically employed to define the MDP. Agent (U) observes the current state $s(l)$ and chooses to act in lth time slot, i.e., $\alpha(l)$. After selecting an action, agent receives the corresponding reward $\varrho(l)$, and then, it observes the next corresponding state $s(l+1)$ in $(l+1)$-time slot. These steps continue until the convergence of training, yielding optimal results.

1) State Space: The state in lth slot is composed of the states of U ($s_U(l) = (\mathbf{q}_U(l), \tau_U(l), \nu(l))$) and IoT devices ($D_n(l) = (E_n(l), A_n(l))$). Specifically, the state of U, denoted $s_U(l)$, consists of its position, time remained for reaching final position, and time-varying velocity, i.e., $\mathbf{q}(l)$, $\tau_U(l)$, and $\nu(l)$ respectively, where $\tau_U(l) = (L-l) - \frac{|\mathbf{q}_F - \mathbf{q}_U(l)|}{\nu(l)}$. The UAV can reach its final location within the mission time T if $\tau_U(l) \geq 0$; otherwise, it fails to do so. The s_n's state in the lth time slot includes its stored energy level and AoI. Among which, the total battery capacity of s_n is divided into ζ energy blocks, with each block containing $\frac{B_n^{\max}}{\zeta}$ joules of energy units. Consequently, the data transfer block and available energy block for EH at the IoT devices in lth time slot are given as $\epsilon_n^T(l) = \frac{\zeta \cdot E_n^T(l)}{B_n^{\max}}$ and $\epsilon_n^H(l) = \frac{\zeta \cdot E_n^H(l)}{B_n^{\max}}$.

2) Action Space: The system's action in the lth time slot represented as $\alpha(l) = \{\alpha(l), b(l)\}$, encompassing the dynamics of U and scheduling of WPT to the IoT devices or data upload to U. In each time slot, U selects a grid to move in the five possible directions. Specifically, U's movement $a(l)$ is given as $a(l) \in \{\mathbb{N}, \mathbb{S}, \mathbb{E}, \mathbb{W}, \mathbb{H}\}$, which represent the four directions and hover position. Scheduling of the D_n for data upload and EH depends on the battery dynamics of the D_n. The energy level ϵ_n of IoT devices' battery in $(l+1)$, is updated as follows: (i). If no data upload is scheduled ($b(l) = 0$), i.e., ϵ_n increases by $\epsilon_n^H(l)$, up to a maximum of E, (ii). If D_n is scheduled to transmit ($b(l) = k$) and has enough energy, i.e., ϵ_n decreases by $\epsilon_n^T(l)$, (iii). Otherwise, ϵ_n remains same.

3) Reward Function: The formulated MDP problem's reward function aims to minimize the AAoI by jointly optimizing U's trajectory and WPT or data upload scheduling at IoT devices. It consists of two components, i.e., the AoI of the packet collected by U from the IoT device, denoted as $G(l) = -\frac{1}{L}\sum_{l=1}^{L} A_n(l)$ and the remaining time to reach the final position ($\mathbf{q_F}$), given by $\tau_U(l)$. The agent should receive an extra reward for reaching the final location on time, while a penalty should be imposed if U fails to arrive. The reward function is expressed as

$$\varrho_{\tau a}(l) = \begin{cases} l(\lambda_1 \tau_U(l) + \lambda_2 G(l)), & \tau_U(l) < 0, \\ l(\lambda_1 \tau_U(l) + \lambda_2 G(l) + \lambda_3), & \text{otherwise}, \end{cases} \tag{6}$$

where λ_1, λ_2, λ_3 are non-negative constants. Since D_n can be scheduled multiple times to lower the AoI of its data, it requires additional energy for numerous transmissions. To minimize overall energy consumption, D_n incurs a penalty

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for energy usage with each data upload, which is denoted as $e(l) = \sum_{n=1}^{N}(e_n(l) - e_n(l-1))$, where e_n is the D_n's battery dynamics. An energy-aware reward is introduced to promote energy efficiency, which is given as

$$\varrho_{\tau ae}(l) = \begin{cases} l(\lambda_1 \tau_U(l) + \lambda_2 G(l)), & l \geq \psi, \tau_U(l) < 0, \\ \lambda_2 G(l) + l\lambda_4 e(l), & l < \psi, \tau_U(l) < 0, \\ l(\lambda_1 \tau_U(l) + \lambda_2 G(l) + \lambda_3), & l \geq \psi, \tau_U(l) \geq 0, \\ \lambda_2 G(l) + l\lambda_4 e(l) + \lambda_5, & l < \psi, \tau_U(l) \geq 0, \end{cases} \tag{7}$$

where $\psi = N - \lambda_5 N$ and λ_4, λ_5 are positive constants.

B. D3QN Framework

The action-value function, $Q^{\pi}(s, \alpha)$, represents the accumulated reward ϱ when taking action α in state s and following policy π that is determined by evaluating $Q^*(s, \alpha) = \max_{\pi} Q^{\pi}(s, \alpha)$. Followed by updating the action-value function iteratively as $Q(s(l), \alpha(l)) = Q(s(l), \alpha(l)) + \gamma(\varrho(l) + \max_{\acute{\alpha}} Q(s(l+1), \acute{\alpha}) - Q(s(l), \alpha(l)))$, where γ denotes the learning rate. In deep Q-networks (DQN), a deep neural network (DNN) having weights, denoted as ω, is used to approximate $Q(s(l), \alpha(l))$ as $Q(s(l), \alpha(l); \omega)$. While training, ω of the current network are updated to obtain the minimum loss function in each time slot, given as $\mathcal{L}(\omega) = (z^{\text{DQN}}(l) - Q(s(l), \alpha(l); \omega))^2$, where the target value is obtained as $z^{\text{DQN}}(l) = \varrho(l) + \max_{\acute{\alpha}} Q(s(l+1), \acute{\alpha}; \acute{\omega})$ with $\acute{\omega}$ as the target network's weights. The D3QN algorithm, described in Algorithm 1, can address the overestimation of the action-value function [12]. Here, a single-stream DNN is divided into two streams for evaluating the value function, $\mathcal{V}_1(s; \omega, \Phi)$, and the advantage value function, $\mathcal{V}_2(s, \alpha; \omega, \Psi)$, the weights of value and advantage function subnetworks are denoted by Φ and Ψ, respectively. The modified action-value function is then given by $Q(s, \alpha; \omega, \Phi, \Psi) = \mathcal{V}_1(s; \omega, \Phi) + (\mathcal{V}_2(s, \alpha; \omega, \Psi) - \Delta\mathcal{V}_2)$, where the size of the action space is given by $\Delta\mathcal{V}_2 = \frac{1}{\mathcal{A}} \sum \mathcal{V}_2(s, \alpha; \omega, \Psi)$ and $\mathcal{A}$. Finally, D3QN algorithm's loss function and $z^{\text{D3QN}}(l)$ is acquired as

$$\mathcal{L}(\omega, \Phi, \Psi) = \left(z^{\text{D3QN}}(l) - Q(s(l), \alpha(l); \omega, \Phi, \Psi)\right)^2, \tag{8a}$$

$$z^{\text{D3QN}}(l) = \varrho(l) + Q(s(l+1), \max_{\acute{a}} Q(s(l+1), \acute{a}; \omega, \Phi, \Psi); \acute{\omega}, \acute{\Phi}, \acute{\Psi}). \tag{8b}$$

The computational complexity of fully connected neural network is expressed as $O(\sum_{n=1}^{N} k_n k_{n-1})$, where k_n is the number of units in the nth layer and N is the total number of layers. Additionally, the complexity of U's exploration and learning process is given as $O(\mathcal{S}\mathcal{A})$, where $\mathcal{S}$ is the number of states that U can occupy and $\mathcal{A}$ is action space size.

V. Simulation Results and Discussions

The simulation is conducted in TensorFlow 2.0 by utilizing two neural networks, each with fully connected hidden layers containing 200 and 256 neurons, respectively. The model uses the Adam optimizer and ReLU as the activation function. The simulation parameters, i.e., episode index $E = 3000$, replay memory $\mathcal{R} = 20000$, and the mini-batch size to 200. The learning rate is 0.00015, with target network updates every

Algorithm 1: Proposed D3QN-Based Data Collection

Initialize U's trajectory, $\mathcal{R} = 20,000$, $E = 3,000$, set $(\acute{\omega}, \acute{\Phi}, \acute{\Psi}) = (\omega, \Phi, \Psi) \in [-1.5, 1.5]$, $P_U = 10$ dBm
for episode $= 1$ to E **do**
1) Initialize environment and observe initial state $s(1)$.
for $l = 1$ to L **do**
2) With probability ρ, select a random action $\alpha(l)$; **otherwise**, choose $\alpha(l) = \max_{\alpha} Q(s(l), \alpha; \omega, \Phi, \Psi)$.
3) Execute action $\alpha(l)$, receive reward $\varrho(l)$ and observe the next state $s(l+1)$.
4) If $s(l+1)$ is a terminal state, store it in memory. Save the transition $(s(l), \alpha(l), \varrho(l), s(l+1))$ in $\mathcal{R}$.
5) Sample a random mini-batch of transitions $(s(b), \alpha(b), \varrho(b), s(b+1))$ from $\mathcal{R}$.
6) Calculate the target value $z^{\text{D3QN}}(b)$.
7) *If* $s(b+1)$ is a terminal state, set $z^{\text{D3QN}}(b) = \varrho(b)$; **otherwise**, calculate $z^{\text{D3QN}}(b)$ using (8b).
8) Update by minimizing the loss $\mathcal{L}(\omega, \Phi, \Psi)$ using (8a).
9) Update target network parameters $(\acute{\omega}, \acute{\Phi}, \acute{\Psi})$.
10) If $s(l+1)$ is a terminal state, then terminate.
end
end

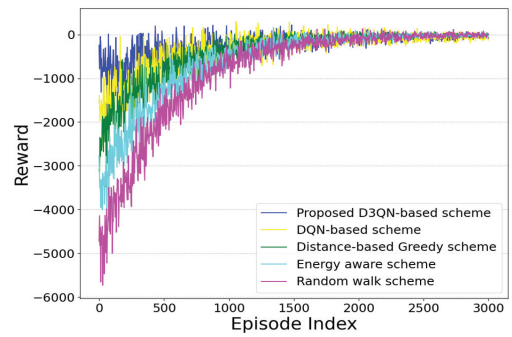

Fig. 1. Convergence behavior of proposed and baseline schemes.

300 steps. U starts from $\mathbf{q}_0 = [-50, -50]$ to $\mathbf{q}_F = [50, 50]$ at $\nu_U^{\max} = 30\,\text{m/s}$ and $H = 70\,\text{m}$. U's parameters are $S_{\text{tip}} = 120\,m/s$, $P_0 = 99.66\,W$, $P_i = 102\,W$, $v_0 = 0.02\,m/s$, $\mu = 1.225\ \text{kg/m}^3$, $A = 0.5\ \text{m}^2$, $s = 0.0001$, and $d_0 = 0.48$. Moreover, $T = 40$ sec, $\beta_0 = -60\,\text{dB}$, $N_U = -60$ dBm, $P_U = 10$ dBm, $\eta = 0.8$, $B = 4\,\text{MHz}$, $L_n = 1$ Mb, $A_1 = 11.95$ and $A_2 = 0.14$.

We evaluate the proposed D3QN scheme's performance against several benchmark schemes to demonstrate its effectiveness. The DQN-based scheme can be trained by using $\mathcal{L}(\omega)$ and $z^{\text{DQN}}(l)$ as described in Section IV-B. Then, a distance-based Greedy scheme, wherein U takes the optimal action based on the current state, adhering to the time constraint in (6). In the energy-aware scheme, U, adhering to the energy-aware reward given in (7), moves to D_n for charging using minimum energy. Lastly, for the random walk scheme, U randomly chooses D_n to update its status and moves in any direction by following step 2 in Algorithm 1. The reward function given to the DQN-based, distance-based Greedy, and random walk schemes is shown in (6). Fig. 1 illustrates the reward progression over 3000 training episodes for the proposed D3QN and benchmark schemes. The rewards fluctuate due to the DRL approach's inherent instability as it explores new actions and interacts with the environment

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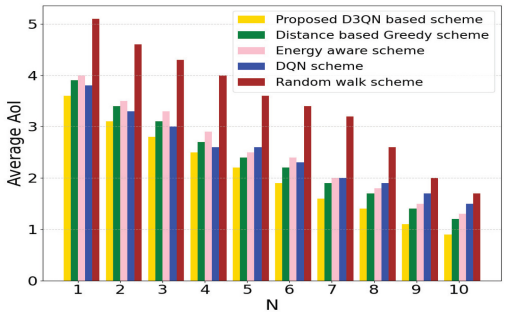

Fig. 2. Average AoI for proposed and baseline schemes.

at a learning rate of 0.00015. Despite these fluctuations, all considered schemes tend to converge to a fixed reward value 1. Additionally, the proposed scheme typically converges faster than DQN because it reduces overestimation bias and provides better target value estimation.

Fig. 2 compares the proposed scheme with benchmark schemes, illustrating the average AoI for different numbers of IoT devices. The proposed D3QN scheme consistently achieves a lower AoI than the baseline schemes across all IoT device counts. The Greedy scheme has a relatively low AAoI, as it selects the best immediate action, though it may overlook energy constraints. The energy-aware scheme prioritizes charging IoT devices with the least energy and achieves moderate AoI, focusing more on charging rather than frequent updates. The energy-aware scheme outperforms the random walk scheme when IoT device numbers are low, moving closer to IoT devices with lower energy. The performance of the DQN-based scheme declines with an increasing number of IoT devices due to state-space explosion, which makes it harder for DQN to learn and optimize policies effectively. Lastly, the random walk scheme yields the highest AoI due to random movement and non-optimized updates, leading to inefficiency. Followed by Fig 3, which shows the optimized trajectories of UAVs around the IoT devices for the proposed and benchmark schemes. In the proposed scheme, U schedules each IoT device to minimize AAoI while maintaining efficient energy use. The DQN scheme displays a path similar to that of D3QN but with extra path length, resulting in higher AoI. However, the distance-based Greedy scheme also creates a relatively efficient path but overlooks the energy constraints. The energy-aware scheme follows a trajectory focused on specific IoT devices for charging rather than optimizing AoI.

VI. Conclusion

This letter addressed the challenges of the limited battery life of IoT devices and the data freshness in UAV-assisted IoT networks, where the UAV wirelessly charges multiple IoT devices using WPT and collects the sensed data from them. We formulated an AAoI minimization problem as an MDP and proposed a D3QN-based algorithm to jointly optimize the UAV's trajectory and scheduling of IoT device for data upload or UAV-assisted WPT. Lastly, the simulation results demonstrated the effectiveness of the proposed scheme. The proposed framework can be applied to various real-world applications, such as smart agriculture, which requires precision farming and resource management and disaster response, which requires timely data collection for situational awareness.

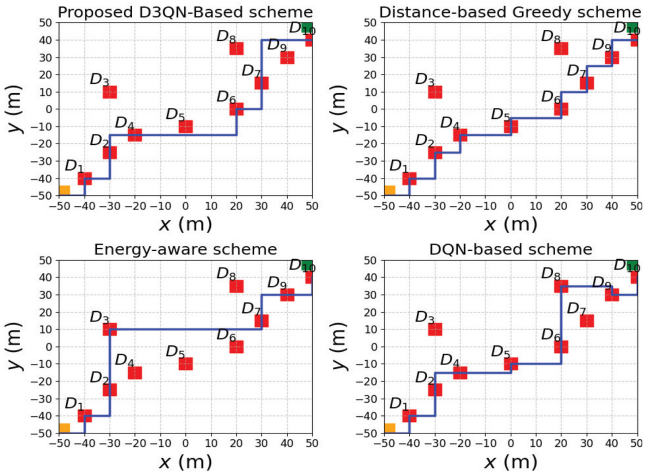

Fig. 3. UAVs' optimized trajectories for proposed and baseline schemes.

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